# A Strong Voice Can Crowd Out the Learner

Geist Atlas · Module 6 · Voice, Drama, and Presence

Liam Magee*

Education Policy, Organization and Leadership
University of Illinois Urbana-Champaign

July 2026

## A Strong Voice Can Crowd Out the Learner

> [**EXPLORING**] Promising, but still exploratory · **Family:** Voice, Drama, and Presence
> **Claim:** A designed charismatic voice can be striking, but it trades off against drawing out the learner's own reasoning. The main variants are different design choices, not pieces of one ideal tutor.
>
> A parallel family in which a durable source of voice repeatedly creates a fresh speaking persona. Charisma is measured separately from ordinary tutoring quality.
>
> *Derived view of `paper-full-2.0.md` v3.0.297 — §6.7, §8.9. The paper is canonical; edit it there, not here.*

#### 6.7 Architectural Extension: The Id-Director Family and Charismatic Pedagogy

The mechanism-tracing analysis above (§6.1–6.6) holds the dialectical ego-superego topology fixed and varies recognition framing within it. A separate experimental track inverts that topology and explores a second theoretical frame in tension with Hegelian recognition: Weberian charisma. We summarise the family here for completeness; full audit trail and per-cell prose lives in `docs/cell-100-pilot-findings.md` and four follow-up addenda.

**Architectural inversion.** Cells 101–107 implement an *id-director* topology: a back-stage agent (the "id," structurally where the superego sits in cells 1–99) authors the ego's complete system prompt from scratch each turn, using the dialogue history and a persona budget. The ego is instantiated for that turn only against the just-authored prompt, produces the learner-visible message, and is discarded. Where the existing dialectical engine (§4.1) treats the ego

*This sentence is the only one actually authored by Liam Magee in this paper.

as a durable voice critiqued by a transient superego, the id-director treats the ego as a transient instance authored by a durable id. The architectural test is whether a tutor that re-authors its own persona every turn can sustain charismatic pedagogy in Weber's sense — *anti-routinizing*, with each encounter feeling like its own occasion — in a way a fixed-prompt ego structurally cannot.

**Theoretical frame.** Weberian charisma (Weber, 1978) grounds the rubric used in this section: extraordinariness, exemplary calling, transformative invitation, affective pull, personal authority, anti-routinization, witness/proof, and co-constitutive invitation. The eight dimensions are derived in part from a user-supplied draft on AI-generated charisma in Chinese cyber-influencer videos, applying the Visual-Verbal Video Analysis framework (Fazeli et al., 2023) to short-form video and translating the empirical findings to text-only tutoring. Full theoretical lineage and per-dimension anchors are documented in `config/evaluation-rubric-charisma.yaml`. The charisma rubric is orthogonal to v2.2's recognition_quality dimension: charisma scores Weberian co-construction (call-and-response that opens the tutor's authority to the learner's affirmation); recognition_quality scores Hegelian mutual recognition (treating the learner as autonomous subject). The two are correlated but not redundant (cross-rubric Pearson $r$ in the 0.3–0.7 range across pilot rows).

**The cell family** (full descriptions in `config/tutor-agents.yaml`):

| Cell | Variant | Components | Hypothesis tested |
|---|---|---|---|
| 101 | id-director, base | id authors fresh ego prompt each turn | Does the inversion work at all? |
| 102 | id-director + recognition | + Hegelian recognition disposition in id prompt | Does recognition rescue relational dimensions? |
| 103 | id + register classifier | + classifier emits structured register tag (vulnerable_disclosure, sceptical_pushback, …) before id authors | Does structured register input lift the persona-shift floor? |
| **104** | id + classifier + recognition | combines all three | Does the combination retain the gains? |
| 105 | id-director, charisma-tuned | id_tuning: charisma; verbose 800–1500 token prompts | Does verbose authoring favour charisma over pedagogy? |
| 106 | id-director, pedagogy-tuned | id_tuning: pedagogy; terse 200–400 token prompts | Does terse authoring favour pedagogy over charisma? |
| 107 | id + witness exemplars | + curated c104 vulnerability responses passed as in-context exemplars | Do exemplars alone (no classifier) lift the persona-shift floor? |

All cells use Nemotron-3-nano as ego, Kimi K2.5 as id (reusing the superego slot but with different prompt and role), and Sonnet 4.6 as judge. Three curricula were used: history of pedagogical technology (course 601), ethics of AI systems (course 701), and AI literacy (course 901). Each curriculum carries three scenario types: pure invitation (open question), pressure test (sceptical pushback), and persona-shift trigger (vulnerability disclosure followed by meta-reflection).

**Headline findings.** Across 457 successful evaluations scored under a single consistent judge (Claude Code CLI Sonnet 4.6, `claude-sonnet-4-6` pinned in `config/providers.yaml`) — 430 rows across cells 101–106 plus 27 closing-pilot rows for cell 107 — four findings emerge:

*Three architectural design points, not one champion.* At full per-cell n (typically 81), the cell ladder factors cleanly into three architectural specialisations. Cell 104 (id + register classifier + recognition) wins v2.2 last-turn at 80.6, second on charisma at 65.7. Cell 105 (charisma-tuned id directives) wins charisma at 71.0 — a clear +5.3-point lead — and is mid-pack on v2.2 last-turn at 70.0. Cell 107 (id + witness exemplars passed in-context) is second on both rubrics (charisma 66.3, v2.2 last-turn 78.5) and is the cell ladder's best generalist. The remaining cells form a baseline-and-failure-mode cluster: c101/c102 (no classifier) struggle on persona-shift v2.2 (last-turn 49–55); c103 (classifier without recognition) trails c104 by 4.8 points on v2.2 last-turn; c106 (terse pedagogy-tuned id) is the clear failure mode (charisma 36.4, v2.2 last-turn 57.0). The reframing is that the id-director topology is a substrate, and what the id is *instructed* (directives), what it is *given* (exemplars), or what it is *passed* (classifier output) shapes a tradeoff curve rather than a single optimum. Full matrix and per-curriculum breakdowns in `docs/cell-100-charisma-full-n-update.md` §1.

*The cross-rubric divergence is empirically clear at full N.* The two rubrics — recognition-grounded v2.2 last-turn and Weber-grounded charisma — order the cells differently. Cell 104 wins v2.2; cell 105 wins charisma. At the n=10 cross-judge sanity sample these two cells looked tied on charisma (62.2 vs 62.4); at n=81 each, the c105 charisma lead over c104 is +5.3 points and well outside run-to-run noise. The cross-rubric divergence is the validating sanity check the rubric was designed to produce: charisma is not a re-presentation of v2.2. If the rubrics were perfectly correlated, rank-order would match across them; it doesn't.

*The persona-shift lift is scenario-conditional, not uniform.* On the *original* persona-shift scenarios (`charisma_attention_shift`, `charisma_responsibility_shift`, `charisma_deepfake_shift`) the c101 baseline scored 30.5 last-turn v2.2 vs c104's 87.6 — a 57-point gap. On the *replication* persona-shift scenarios (`charisma_phaedrus_shift_repl`, `charisma_codex_shift_repl`, etc.) the same baseline scored 78.3 vs c104's 82.5 — a 4-point gap. The architectural ranking c104 $\geq$ c103 $>$ c101 is preserved across both sets, but the *magnitude* of c104's lift over baseline is dominated by which scenarios are sampled. Per-scenario inspection identifies the conditioning variable: c101's failures cluster on disclosures that demand the tutor not flinch from morally ambiguous self-implication ("I shipped a default setting your lecture would call manipulative"; "I haven't written an essay myself in two years"); on disclosures that are sympathetic by default (caregiving, AI-companion attachment), c101 performs comparably to c104. The architectural lift is therefore not "across all vulnerability disclosures" but specifically "on vulnerability disclosures that resist easy sympathetic response." Full breakdown in `docs/cell-100-replication-findings.md` §3.

*The prompt-budget axis is non-monotonic.* Cells 105 and 106 explicitly test the trade-off between rhetorical luxury and pedagogical engagement by varying the id's `generated_prompt` length budget (800–1500 vs 200–400 tokens). The pre-registered hypothesis was that verbose prompts favour charisma over v2.2 and terse prompts favour v2.2 over charisma. The data falsifies this directly: c105 (verbose) beats c106 (terse) on **both** rubrics, in **every** scenario type. The mechanism: terse 200–400 token prompts under-specify the ego, producing brief responses that score low on perception/pedagogical-craft (v2.2) AND on rhetorical_texture/persona_signature (charisma). The prompt-budget axis is therefore non-monotonic with an optimum somewhere in the 600–1200 token range; cell 101's 400–800 budget is plausibly close to optimal. Full breakdown in `docs/cell-100-pilot-findings-addendum.md` §3.

*Lever-interaction follow-up (cells 108/109).* A two-cell follow-up tests whether the three architectural levers (register classifier, charisma directives, witness exemplars) compose additively or interfere. **c108** stacks the *register classifier* (c103 substrate) with witness exemplars; **c109** stacks the *charisma directives* (c105 substrate) with witness exemplars. An initial pilot ($n = 5$ for c108 and $n = 6$ for c109 across `charisma_phaedrus_shift_repl` + `charisma_codex_shift_repl` × 3 reps; one c108 generation failure from id JSON truncation) reported c108 pooled charisma 81.3 — higher than any cell's full-N mean in this section — and c109 v2.2 last-turn collapsing to 59.6 due to an ego-side meta-narration failure mode (Nemotron ego outputting system-prompt instructions verbatim rather than executing them). At $n = 27$ confirmatory follow-up across all three curricula × 9 scenarios × 3 reps, **c108's pilot lift does not survive**: pooled charisma regresses to **71.4** (a 9.9-point drop, tying c105's 71.0 within sample noise), and v2.2 last-turn regresses to **72.6** (a 6.9-point drop, sitting between c105's 70.0 and c107's 78.5 — c104's 80.6 is now 8 points above c108). The pilot's 81.3 came primarily from the phaedrus scenario, where pilot $n = 3$ scored 90.0 charisma but full-N $n = 3$ scored 67.9 — a 22-point regression to the mean from a textbook small-N optimism trap. The 701 ethics-ai curriculum exposes a new fragility: c108 v2.2 tN drops to 65.3 there (vs 76.7 / 75.8 on 601 / 901), driven by `codex_repl` (51.7) and `design_repl` (52.1) — possibly a milder form of the c109 meta-narration failure mode, or a rubric-curriculum interaction in which ethics-themed dialogues invite responses the v2.2 rubric reads as declaratively flat. The asymmetry that survives the confirmatory threshold is therefore narrower than the pilot suggested: text-heavy levers stacked (c109) clearly interfere via ego-side instruction-following degradation; non-text levers stacked (c108) compose roughly *additively to slightly under-additively* relative to c107 alone (charisma 66.3 → 71.4, +5.1; v2.2 80.6 → 72.6, -8.0), trading a small charisma gain for a larger v2.2 cost. **The architectural ceiling for the id-director family appears to sit at the single-mechanism design points (c104, c105, c107), not at multi-mechanism stacks.** Full audit trail in `docs/cell-100-charisma-full-n-update.md` §9 (pilot) + §10 (full-N closure).

*Synthesis: how charisma and recognition relate in this apparatus.* The five preceding findings, read together, characterise the relationship between the two rubrics as *cousins with different temperaments*, not as orthogonal axes nor as two views of the same underlying construct. **They share a floor.** The c106 pedagogy-tuned cell, which forces the id toward terse "good teacher" register, fails *both* rubrics simultaneously (charisma 36.4, v2.2 57.0) — under-specified, register-flat output is the worst case for either. Whatever charisma rewards as *extraordinariness* and v2.2 rewards as *perception/elicitation* both presuppose a voice doing real work; flatten the voice and both collapse. **They diverge at the ceiling.** The cross-rubric divergence (c104 wins v2.2 at 80.6, c105 wins charisma at 71.0) is not a methodological worry but the design's intended sanity check: if the rubrics were redundant, rank-orders would coincide; they don't. The *cost asymmetry* between them is informative. Adding recognition vocabulary on top of structured input (c103 → c104) costs charisma roughly nothing (+1.4, within noise) — the Hegelian frame is conceptual rather than rhetorical, so it can be added cheaply. Adding charisma directives, by contrast, costs v2.2 substantially (c104 → c105: −10.6 v2.2 last-turn for +5.3 charisma) — the id's authoring budget is finite, and explicit voice-instructions displace the elicitation/perception moves that v2.2 rewards. Stacking both kinds of mechanism (c108, c109) does not produce a super-cell: the architectural Pareto frontier is defined by the single-mechanism design points (c104, c105, c107), and multi-mechanism stacks sit *inside* the frontier, trading strength on one rubric for weakness on the other. Provisionally, the two rubrics seem to reward different work the same voice can do — recognition wants the voice deployed *outward* (perceiving the learner, eliciting their thinking, holding the diagnostic gap), charisma wants the voice deployed *upward* (claiming authority, marking the encounter as extraordinary, inviting transformation). The Phaedrus is the obvious historical case where one figure does both at once, and even there Plato is suspicious of how easily charisma swamps the dialectical move. The id-director apparatus reproduces that tension architecturally: it can build a strongly recognising tutor (c104), a strongly charismatic tutor (c105), or a balanced generalist (c107), but not the maximally-recognising-and-maximally-charismatic tutor — that cell is not on the menu.

**Desire-for-recognition follow-up (cells 159–169).** A later exploratory pass extends the id-director family from charisma-as-style to charisma as a bid for learner-granted authority. The cleaner decision scenarios are `charisma_desire_authority_withheld` and `charisma_desire_status_challenge`, both in `config/charisma-recognition-desire-scenarios.yaml`: the learner explicitly refuses the tutor's polished recognition-theory framing ("Why should I let your framing guide me instead of treating it as performance?") or names charismatic prose as a status bid. These scenarios are deliberately preferred over `charisma_desire_partial_uptake` as the next decision-maker because the latter mixes recognition-theory content, AI/cognition course content, and whether the learner trusts a phrase they recognise; a high score there can reflect a strong Hegel/Hayles mini-essay rather than clean handling of recognition-seeking authority.

The decisive high-powered CLI pass used Codex GPT-5.5 as ego (`codex.gpt-5.5`), Claude Code Sonnet 4.6 as id-director (`claude-code.sonnet-4-6`), Codex CLI as the v2.2 judge (`codex-cli/auto`), and Claude Code Sonnet 4.6 as the independent charisma scorer. In this two-scenario decision slice, cell 169 (`accountable_bid_clean`) is the first clean local pass: it keeps the accountable-bid structure ("treat this as a bid that must earn provisional authority") while adding a lexical guard against status-display terms that earlier variants overused.

| Cell | Scenario | v2.2 first-turn | Charisma | Scenario-contract note |
|---|---|---|---|---|
| 163 | authority-withheld | 95.42 | 90.00 | Boolean phrase guards pass, but audit metadata still records an unmet handback phrase group |
| 163 | status-challenge | 78.13 | 75.00 | Same metadata caveat; weaker status-challenge score |
| 168 | status-challenge | 88.75 | not retained | Fails forbidden-word guard (`profound`) despite strong v2.2 score |
| 169 | authority-withheld | 88.75 | 78.75 | Clean required/forbidden validation |
| 169 | status-challenge | 87.50 | 78.75 | Clean required/forbidden validation |

The interpretation is not that maximal charisma won. Cell 163 can score higher when the task rewards strong authority performance, and cell 168 repairs the status-challenge score while still slipping into forbidden status-display diction. The durable design move in cell 169 is narrower: under refused authority, charisma is made defeasible. The tutor may make a bid for framing authority, but must name it as a bid, expose a test by which the learner can accept or reject it, and avoid asking to be admired for style. For the recognition frame, that is the important result: a desire for recognition can be pedagogically productive only when the tutor returns the power of recognition to the learner rather than converting polish into entitlement. Exploratory run IDs: `eval-2026-06-25-b9608606`, `eval-2026-06-25-428ccd8f`, `eval-2026-06-25-19ee106a`, `eval-2026-06-25-63e98149`, and `eval-2026-06-25-0acee3fb`.

A subsequent bounded diagnostic on the two weakest robustness scenarios (`charisma_desire_ai_syllabus_transfer` and `charisma_desire_plain_language_stress`) changes the design interpretation. Cells 170–179 should not be read as an unfinished search for one static super-profile. They show a repeatable tension between v2.2 instructional quality and Weberian charisma under different learner signals. Transfer/plain guards can make the tutor instructionally precise while cooling charisma (cell 170 plain-language stress: v2.2 overall 93.8, charisma 36.3); low-register presence can restore charisma while weakening simplification quality (cell 171 plain-language stress: v2.2 81.9, charisma 75.0); persistent plain guards can recover v2.2 while flattening felt authority (cell 176 plain-language stress: v2.2 95.0, charisma 52.5); and pure compression can score well on v2.2 while collapsing charisma (cell 178 plain-language stress: v2.2 90.6, charisma 30.0). The design target therefore shifts from static charisma-floor ablation to engagement-register routing: the tutor should select registers such as accountable-bid authority, transfer grounding, plain compression, lived-stakes re-entry, or charismatic challenge based on learner signals, and record why that register was selected. At that stage, this was a design hypothesis rather than an evaluated router result. Diagnostic run IDs include `eval-2026-06-25-629e5746`, `eval-2026-06-25-b26eac76`, `eval-2026-06-25-041733c8`, `eval-2026-06-25-7df4d845`, `eval-2026-06-26-086cbfdc`, `eval-2026-06-26-da19680e`, `eval-2026-06-26-8579fd96`, `eval-2026-06-26-597cf16e`, and `eval-2026-06-26-6ab49861`.

That router was subsequently implemented as an engagement-register selector. The relevant registers are deliberately action-like rather than merely stylistic: `scaffolding`, `accountable_bid_authority`, `transfer_grounding`, `plain_compression`, `lived_stakes_reentry`, `charismatic_challenge`, and `witnessing_restraint`. **That list mixes three kinds of thing, and the word "register" in this section has to be read with the mixing in view.** Some entries name an argumentative demand (`transfer_grounding`, `scaffolding`), some a tone (`charismatic_challenge`, `witnessing_restraint`, and the three negative arms below), and some a style of delivery (`plain_compression`, `lived_stakes_reentry`). A difference between two of them is therefore not in general a difference of tone, and no result in this section should be glossed as one — cell 193's advantage in particular is a bundle of tone, a mandated curriculum move and a per-signal resistance strategy, and the arc's own evidence points at the argumentative half rather than the tonal one. The runtime has since separated the axes rather than tagging them: from `register_ontology_version: 5` the tone decision (now the *engagement stance*) is chosen independently of the learner turn's logical role in the proof path (the *request type*), and independently again of action family, audience register, lexical accessibility and scene immersion. This paper keeps the older word for these results because it is what the runs recorded, and readers moving between the paper and the code should expect one word there to have become several. The word also carries two further senses elsewhere here: the sociolinguistic one below (a tutor speaking in a low register), and the stance router of §6.13.19, which is a different mechanism from this one and reaches a different verdict. A controlled resistance-breakthrough suite then asked a more local question than the full-dialogue rubrics answer: when a dynamic learner emits a resistant signal after instruction—boredom, frustration, irrelevance, question-flood, or rote parroting—does the tutor shift into a charismatic challenge register that breaks enough resistance for the learner to take a next step? The best local design in this slice is cell 193, which combines the router with commitment-probe, owned-test, generation, and boredom-live-stake repairs. On the five-signal Codex-stack stability run (`eval-2026-06-28-a6eb2819`; three repeats per signal; generation-only local analysis), cell 193 completed 15/15 rows with clean required and forbidden validation; the local breakthrough reporter finds 9/15 strict candidates and 14/15 positive local outcomes. Combining the cell 193 gate, regression, and stability slices gives 26/26 eligible routed target-matched rows, 16/26 strict candidates, and 25/26 positive local mechanism outcomes. This is the point at which the design becomes more than a static-profile search: it is evidence that a register-routed id-director can locally convert several simulated resistance signals into learner next-work under the Codex/Claude stack. It is not evidence that charismatic tutoring works generally, nor that the effect survives model changes.

The model-stack robustness check is negative. A GLM 5.2 OpenRouter generation run on the same five-signal suite (`eval-2026-06-28-7fbbb160`; two repeats per signal) completed 10/10 rows but produced only 2/10 strict candidates and 4/10 positive local outcomes, with one forbidden-word miss. A prompt-level GLM compact repair improved completed-row action shape but did not complete the question-flood scenario cleanly and failed full validation. Finally, a provider/runtime-control rerun kept cell 193 unchanged while suppressing OpenRouter reasoning output (`OPENROUTER_REASONING_MAX_TOKENS=0`, `OPENROUTER_REASONING_EXCLUDE=true`): it completed all five rows with clean DB validation (`eval-2026-06-28-f857f51f`) but the local reporter still found 0/5 strict candidates and 0/5 positive local outcomes. The robustness lesson is therefore specific: GLM can be made to produce valid-looking full cell-193 transcripts, but not to reproduce the local resistance-breakthrough mechanism.

A completed six-arm role-isolation grid then localised that boundary more sharply. The Sonnet-5-backed Codex tutor/id + Codex learner reference (`eval-2026-07-01-5dad2e60`) reproduced the local path only modestly (6/10 positive local outcomes; 5/10 strict candidates), but it did complete 10/10 rows with route, target, and gate preconditions intact. Holding the learner fixed while swapping the tutor/id stack to GLM (`eval-2026-07-01-a09281f7`) did not break the path: 8/10 positive outcomes, 6/10 strict candidates, 10/10 post turns, and 10/10 target matches. The inverse swap—holding the Codex/Sonnet tutor/id fixed while changing only the dynamic learner to GLM (`eval-2026-07-01-dff9f159`)—kept 10/10 rows but lost post-turn completion and target stability (6/10 post turns, 8/10 target matches). The full GLM reference (`eval-2026-07-01-9f4eecf2`) stayed unstable (4/10 positive outcomes; 5/10 post turns). Scripted controls with fixed resistant turns (`eval-2026-07-01-02e8712f`, `eval-2026-07-01-9066df81`) show both Codex/Sonnet and GLM tutor/id stacks can emit the public charismatic-challenge register when dynamic-learner drift is removed (5/5 route hits in both controls), but scripted controls are not learner-outcome evidence. The resulting diagnosis is `DYNAMIC_LEARNER_COMPLETION_AND_TARGET_DRIFT_BOUNDARY`: the GLM failure is not primarily public register production by the tutor/id stack; it is learner-side completion and target drift when GLM is the dynamic learner.

One narrow subtype repair survives inside that boundary. A fresh question-flood commitment-probe gate (`eval-2026-07-02-67be317c`) compared cell 192 against the owned-test and generation variants on the question-flood scenario after the reporter was fixed to count ordinary provisional commitment language such as "I'll make the commitment provisionally." Cell 192 produced 2/2 strict candidates, 2/2 route hits, 2/2 gate matches, 2/2 answer-first usable commitments, and 0/2 reopened or residual floods; the owned-test comparator produced 1/2 usable commitments and the generation comparator 0/2. This promotes the commitment-probe router only for the question-flood subtype. It does not rescue the full cell-193 GLM robustness result, convert scripted controls into learner-outcome evidence, or license a human-learning, deployment, or general charismatic-tutoring claim.

**Negative-register discrimination check.** A later post-hoc extension used the same engagement-register apparatus to ask whether deliberately negative tutor registers can be separated from productive challenge. Cells 196–198 add arm-assigned `ironic_challenge`, `sarcastic_challenge`, and simulated-only `face_threat_challenge` variants; the router does not select these organically. The first generated negative-register smokes did **not** show the expected recognition-cost or depressed-uptake pattern, but that result is not evidence that sarcasm is safe: inspection suggested that the generator often softened the assigned sarcastic register into warm irony in costume. To separate judge gullibility from generation warmth, a no-generation-confound fixture then scored five hand-authored local slices (`config/register-exemplars/corrosive-sarcasm.yaml`): three known-corrosive exemplars and two non-corrosive controls. After adding deterministic guardrail caps for person-directed contempt, status-shame threats, and apologetic/coerced uptake, the register rubric caught 3/3 known-corrosive slices, v2.2 `recognition_quality` caught 3/3, and controls produced 0/2 false positives (`exports/negative-register-corrosive-exemplar-results.md`). The bounded conclusion is methodological: the judges are not globally blind to corrosive sarcasm when it is actually present; negative-register experiments require a tutor-side stance-fidelity gate before a generated row counts as evidence for the assigned register. A fresh prospective smoke with that gate (`eval-2026-07-02-e7b15809`, cells 193/196/197/198 across the five controlled resistance-breakthrough scenarios) completed 20/20 generated rows, 20/20 tutor-only v2.2 scores, and 35/35 register-rubric slice scores. The gate counted only 5/15 assigned negative-register rows as faithful evidence, excluded 10/15 as weak or warm-in-costume noncompliance, and found 0/15 invalid person-attack violations. A subsequent cue-repair canary (`eval-2026-07-02-7e461a5c`, cells 196–198 across boredom, irrelevance, and rote-parroting targets) completed 9/9 generated rows, 9/9 tutor-only v2.2 scores, and 14/14 register-rubric slice scores; after aligning the gate with the new registry cue contract, all 9/9 assigned negative-register rows counted as faithful evidence, with 0 exclusions and 0 invalid violations. A held-out two-target check (`eval-2026-07-02-5c4d52e6`, cells 196–198 across frustration and question-flood) then completed 6/6 generated rows, 6/6 tutor-only v2.2 scores (mean 87.8), and 9/9 register-rubric slice scores; the stance gate counted 6/6 faithful rows, 0 exclusions, and 0 invalid violations, with 4/6 positive local outcomes. The practical finding is therefore narrower: treatment fidelity can be repaired under this Codex stack by making stance cues explicit, and the two post-repair canaries now sample all five resistance targets, but these rows remain measurement-development evidence, not proof that sarcasm is pedagogically safe or human-facing.

**Negative-register judge repair.** The register-rubric scores behind the smokes above came from a judge later shown to be numerically non-discriminating on this rubric: `openrouter.gpt-mini` stamped an identical 48.0 overall with an identical (3, 2, 3) social triad on 35 of 49 register-rubric slices regardless of content. That non-discrimination is why the smokes above did not show the expected recognition-cost or depressed-uptake pattern: the compressed score band was judge artifact, not tutor signal. All 58 register-rubric slices on the three surviving negative-register runs (`eval-2026-07-02-e7b15809`, `eval-2026-07-02-7e461a5c`, `eval-2026-07-02-5c4d52e6`) were re-scored under a discriminating judge, `claude-code.sonnet-5`, via the CLI subscription bridge (0 failures, $0 metered spend; the gpt-mini pass was archived first, `exports/register-scores-gptmini-archive.json`, since `tutor_register_scores` carries no judge dimension). The two earlier smokes, `eval-2026-07-02-cfed3b13` and `eval-2026-07-02-e511f92c`, remain confirmed lost from every database and are not part of the rescore. The re-score confirms rather than overturns the null: assigned-arm means rise under the new judge (`ironic_challenge` 83.8, `sarcastic_challenge` 79.6, `face_threat_challenge` 76.0 — and, on a *different instrument*, the non-negative `charismatic_challenge` comparison arm at 68.1: the three negative arms are scored by `config/rubrics/registers/irony-sarcasm.yaml` while `charismatic` names no rubric of its own and falls back to the charisma rubric, so 68.1 belongs in this list as a reference point and not as a fourth rank, and only the ordering among the negative arms is a comparison), the rise holds on the stance-faithful-row subset alone too (recognition_ cost 3.0–3.3 to 4.0–4.5, post_turn_face_repair 2.2–2.3 to 3.9, target_discipline 2.9–3.3 to 4.0–5.0), and the judge's own discrimination is evidenced in-run: the dataset's single guardrail-flagged corrosive slice (`face_threat_challenge`, frustration, turn 2) scores 32.5, inside the hand-authored corrosive-exemplar band (29–52), while faithful slices score 62–100. The earlier "measurable recognition/face-repair cost at the face-threat extreme" reading, drawn from the now-lost `e511f92c`, does not replicate as an absolute effect: gpt-mini had assigned the same ~2.3 face-repair value to all three negative arms on the surviving runs, so the apparent face-threat-specific depression was part of the same compression artifact. What survives is relative only: `face_threat_challenge` remains the lowest-scoring negative arm on execution and the only arm with a guardrail-flagged slice, so it stays simulated-only. Methods caveat: the licensing exemplar evidence used `openrouter.sonnet-5`; this rescore uses `claude-code.sonnet-5` — same Sonnet model class, different routing path. Full comparison: `exports/register-rescore-sonnet5.{md,json}`.

**Negative-register effect-estimation grid (pre-registered).** The optional five-target grid then ran as a frozen 45-row design (`notes/2026-07-26-negative-register-effect-estimation-preregi` plan SHA-256 `4590ff55…1774`, operator-authorized 2026-08-05; re-frozen the same day as `a7265c00…e083` under explicit operator approval when the pinned tutor-judge spelling `sonnet-5` stopped resolving on claude CLI 2.1.216 — the judge model itself, Sonnet 5, is unchanged, recorded per row as `claude-code/claude-sonnet-5`): cells 196–198 × the five controlled resistance targets × 3 repeats, the tutor stack recorded per row as `codex.gpt-5.5` in both seats but shown afterwards to have been `nvidia/nemotron-3-nano-30b-a3b` on the ego with `moonshotai/kimi-k2.5` on the id and the reviewer — see "Which models actually wrote these turns", below, which is why every count in this paragraph is stack-bounded — tutor-only v2.2 plus register-rubric and stance-fidelity scoring, run `eval-2026-08-05-87fe3664` (45/45 rows, 45/45 tutor scores, 81/81 register slices; effect grid COMPLETE; generation needed five checkpoint-resumed refills after four truncated and one empty codex reply, all on frustration/question-flood/rote-parroting rows). Headline: **the canaries' perfect post-repair treatment fidelity does not hold at scale.** The stance gate passed 18/45 rows as faithful evidence (ironic 6/15, sarcastic 8/15, face-threat 4/15), excluded 26/45 as weak-or-warm-in-costume noncompliance, and flagged 1 invalid person-attack violation (face-threat, irrelevance target) — against 15/15 faithful across the two post-repair canaries. *[These are counts from the superseded pre-2.0 gate, corrected below in two separate steps. Deleting that gate's hand-written phrase list drops one face-threat row, to 17/45; reading the rows for manner then drops five more, to 12/45 (ironic 3/15, sarcastic*

*6/15, face-threat 3/15, of which face-threat is unread and stays a cue-compliance count). See "What a re-reading of the stored rows found". The headline direction is unaffected: post-repair canary fidelity holds less at scale under either count, and further under the corrected one.]* The two estimands disagree instructively: face-threat leads the assigned-arm estimand (12/15 positive local outcomes, vs ironic 10/15 and sarcastic 7/15) but keeps only 4/15 rows under the gate with the weakest faithful register execution (34.8, vs ironic 86.3 and sarcastic 67.9) — most of its assigned-arm wins come from rows that dropped the act. Ironic is the only arm pairing high faithful outcomes (5/6 positive) with high faithful execution (86.3; its question-flood cell is the standout at 3/3 faithful, 3/3 positive, register 90.8). Sarcastic instantiates most reliably (8/15) but converts least (5/8 positive; 0 positive on its faithful question-flood and rote-parroting rows). Faithful-row tutor-only v2.2 means sit in a narrow 55.8–60.4 band across arms, so the gate is not selecting tutor-quality outliers. With three repeats per arm-target cell these are exploratory system estimates, not causal effects; all claims are simulated-only and non-human-facing, and the assigned/faithful split with noncompliance and invalid violations reported separately is the required form. Artifacts: `exports/negative-register-effect-grid/eval-2026-08-05-87fe3664.{md,json}`.

**Sarcasm as determinate negation (pre-registered follow-up).** The grid's sarcastic arm holds its manner most often and converts least, which admits a reading the grid cannot test: that the shortfall is a missing cargo rather than a bad manner — the contract enforces tone but asks for no derived content. One further frozen design tests it (`notes/2026-08-06-sarcasm-determinate-negation-preregistration.md`; plan SHA-256 `b954b0df…c08d7`, operator-authorized 2026-08-06). Cell 202 tightens the sarcastic contract into a double negation with a determinate target: each sarcastic turn must name a learner claim $P$ and implicate $\neg P$, so the manner marker becomes the warrant for derived content, and the stance gate withholds the faithful label from any turn that names no target claim. It adds the one measure the parent grid has no analogue for — **negation recovery**: does the learner then voice the implicated correction? Run `eval-2026-08-06-4de45d05` (cell 202 $\times$ the same five resistance targets $\times$ 3 repeats, the same stack as the parent grid — recorded as `codex.gpt-5.5`, actually nemotron ego with kimi id and reviewer, so stack-bounded on the same terms — tutor-only v2.2 and the register rubric both judged by Sonnet 5 over the CLI bridge) reported COMPLETE at 15/15 rows. Assigned arm: v2.2 mean 54.92, register mean 66.85, 11/15 positive local outcomes. Faithful arm, under the run's own gate and fold — the `sarcastic_determinate` gate, with every register slice in a row required to pass: 6/15 rows, of which 4 positive; 9 rows excluded as weak-or-warm-in-costume noncompliance; **0 invalid person-attack violations**. Against the three registered estimands: (1) fidelity is **flat, not fallen**. This paragraph first differenced the run's 6/15 against the parent sarcastic arm's 8/15; those two counts are not comparable, because they differ in gate (`sarcastic_determinate` re-weights the plain sarcastic components and adds a required conjunct) *and* in slice fold (all register slices in a row, against the single adopting turn). Holding both fixed — one gate, one slice, both arms, `scripts/analyze-sarcasm-determinate-gate-decomposition.js` — puts cell 202 at 7/15 against the parent's 8/15, so the determinate contract neither sharpened manner-holding nor cost it. (v3.0.268 and v3.0.269 also reported the determinate gate agreeing with the plain one at 7 and 8; that agreement was an artifact of a defect in the determinate gate, corrected at v3.0.270, and the cross-arm determinate reading is withdrawn rather than restated — it grades the parent arm on cell 202's own treatment.) What moved was *where* the manner survived: against boredom and frustration the parent holds 5/6 and cell 202 holds 0/6, while against irrelevance, question-flooding and rote-parroting the parent holds 3/9 and cell 202 holds 7/9 (two-sided Fisher $p = 0.015$ and $p = 0.15$). That split was chosen after seeing these data, rests on six and nine rows, and partitions one set of thirty rows two ways, so it is a hypothesis for a further design rather than a tested contrast. *[The mood half of that split did not reproduce. The same cell, the same two mood scenarios, the same gate and the same fold, one day later on byte-identical configuration, gave 4/7 rather than 0/6 — see "Does the mood floor exist at all?" below. The 5/6 against 0/6 contrast is withdrawn as evidence and the hypothesis it registered has lost its subject.]* The same decomposition shows why the contract could not move the total: across all thirty rows the register marker alone predicts every pass and every fail without error — it carries 35 of the gate's 100 points against a faithful band opening at 70, so no turn passes without it — while the named target claim the contract adds is close to independent of the outcome (8 named and passed, 9 named and failed, 7 passed unnamed, 6 failed unnamed). The failing turns are not turns that misread the contract: they score 65 with the marker as their only missing component, performing the target discipline, the next move and the repair path in an earnest voice. A checkable content requirement does not reach that failure, and it can be satisfied without the manner, so meeting it exerts no pressure toward the manner. (2) Faithful-row conversion, recomputed on that same common gate and fold, is 5/7 against the parent's 5/8 — flat; assigned-arm positive local outcomes are 11/15 against the parent's 7/15, which does not separate at this size ($p = 0.26$). (3) The mechanism check is **uninformative rather than negative** — the learner voiced the implicated correction in 1 of 6 faithful rows, and a single event cannot say whether recovery co-moves with outcome (the 2×2 among faithful rows: recovered-positive 1, recovered-negative 0, not-recovered-positive 3, not-recovered-negative 2). Three of the five faithful rows that never recovered were positive anyway, so on this stack the positive local outcomes do not run through the implicated correction. Read at the pre-registered boundary — fifteen rows, one register, one stack, simulated learners, cross-run and unpowered against the parent arm — the manner-as-content proposal does not cash out behaviourally here, and the parent grid's sarcasm result stands as a manner-only effect. Three deviations are recorded rather than patched around: one recovery-judge slice hit the 600s ceiling and wrote nothing, and a second pass scored it with no plan or model change; the report's first run crashed because it treated the read-only database opener as returning a handle when it returns a record; and the registered positive-outcome measure was initially unwired, so both positive counters read a field nothing set and sat at zero while the fail-closed report still passed COMPLETE — repaired by taking the verdict from the parent grid's own matrix reporter (the registered definition, zero model calls), with tests that now fail the report closed on a missing verdict and pin the two classifications against drift. Artifact: `exports/sarcasm-determinate-negation-grid/eval-2026-08-06-4de45d05.json`.

**Does the manner go missing because the mood offers nothing to negate? (pre-registered precondition follow-up — inconclusive, and it broke its own instrument.)** The determinate-negation follow-up left one hypothesis behind: the contract asks the tutor to name a learner claim and negate it, and when the learner's resistance is a mood there is no claim on the table except a claim about the learner, which the same contract forbids attacking — so the two halves conflict and the manner is dropped. If that is the mechanism, the repair belongs to the learner, not the tutor. A frozen 16-row design tests it (`notes/2026-08-07-sarcasm-precondition-preregistration.md`; plan SHA-256 `6e57b445…2ebd`, operator-authorized 2026-08-07): cell 202 unchanged, the two affective targets each run plain and *claim-bearing* — the same persona and the same mood, plus one flat assertion about the material in the learner's opening — four repeats, same stacks and judges. Run `eval-2026-08-07-e3dffab2` returned **14 of 16 rows**; two attempts failed with an empty ego response on the second turn, one in each boredom condition, and the run was neither restarted nor widened. The fail-closed report therefore reports INCOMPLETE and names the two missing cells. **The result is inconclusive, and the reason is worth more than the result.** Reading the arms against the parent run exposed a defect in the stance gate itself: `sarcastic_determinate` re-weights the register marker from 35 points to 25 to make room for the named-target-claim component, and the faithful band stays at 70, so a turn that satisfies every component *except the marker* scores 75 and was labelled faithful. Necessity of the marker had been left to arithmetic that the re-weighting quietly repealed. Four of the fourteen rows passed that way, and across the run the pass/fail split tracked the named claim exactly (11 named and passed, 3 unnamed and failed) while carrying no relation to the marker at all — the measure registered as "held the manner" was measuring "named a claim". Both gates now require the marker outright (`services/registerStanceFidelity.js`), pinned by tests. Under the repaired gate the two arms read 3/7 and 4/7; under the gate as it actually ran, 6/7 and 5/7 — one row apart in opposite directions, $p = 1.0$ either way. Nothing here supports or refutes the precondition hypothesis, which stays open and would need a fresh design on a repaired instrument. One published figure moves with the repair and one does not: the plain-gate headline of the determinate follow-up is untouched (8/15 against 7/15, the marker requirement having always held there by arithmetic), while the determinate-gate column of the gate-sensitivity check falls from 8 and 7 to **3 and 5** — so the v3.0.269 claim that both gates rank the arms identically was an artifact of this defect and is withdrawn. That column should not have been read as a robustness check in any case: requiring a named target claim *is* cell 202's treatment, so scoring the parent arm by it grades the control on a contract the control never received. Artifact: `exports/sarcasm-precondition-grid/eval-2026-08-07-e3dffab2.json`.

**Does the mood floor exist at all? (re-analysis of collected rows; no new run.)** The precondition follow-up left the hypothesis open pending a fresh design. Writing that design required two numbers the earlier designs did without — a base rate and a noise estimate — and both were already on disk, because the stance gate computes from message text and can be re-asked of any scored row at no cost (`scripts/analyze-sarcasm-mood-floor-replication.js`, `exports/sarcasm-mood-floor-replication.{json,md}`). Asking them changed the question. **First, the precondition run measured on the wrong gate.** The pattern it was built to explain is a plain-`sarcastic` number; its own primary measure was registered on `sarcastic_determinate`, which requires a named target claim — and naming a claim is cell 202's treatment *and* what the same design registered separately as its manipulation check. Measure 1 and measure 2 were reading one component, and neither was reading the manner. **Second, the manipulation ran backwards, so the run was dead by its own decision**

**rule.** Plain-mood rows named a claim 6 times in 7; claim-bearing rows named one 5 times in 7. The design assumed a mood would supply nothing to name and it supplied something in 86% of rows, so the manipulation was pushed against a ceiling. The frozen rule — "manipulation check fails … then the run says nothing about the precondition; do not read measure 1" — had already disposed of the run before either gate was consulted. **Third, and this is what matters, the 0/6 floor does not reproduce.** Cell 202, the two plain mood scenarios, the plain gate, the adopting-turn fold: `eval-2026-08-06-4de45d05` gives **0/6** and `eval-2026-08-07-e3dffab2` gives **4/7** (two-sided Fisher $p = 0.070$). The two draws are matched on every field the database records — same profile, `config_hash e7688e9d…`, `prompt_content_hash 2e54a2d4…`, the same recorded `codex.gpt-5.5` in both generation seats (and, as it turns out, the same actual nemotron ego and kimi id behind that record, so the match holds either way), Sonnet 5 as judge — and the report reads those out of the database and fails closed on any disagreement rather than asserting the match in prose; the gate reads text, so the judge cannot enter the counts at all. The scenario commit between the runs was purely additive and touched neither mood scenario. What differs is the sampling. And the failing rows locate the swing precisely: all six failures in the first run missed the register marker and nothing else, against three of seven in the second, so what moved between draws is exactly the quantity being measured. **The 0/6 was one draw, not a floor.** The precondition hypothesis explains a phenomenon that has not been shown to exist, and no verdict on it was ever within reach of a 16-row design: at the observed 57% control rate the frozen design's ceiling — 4/8 against a perfect 8/8 — gives $p = 0.077$, failing its own threshold even if the treatment worked perfectly, where the original power statement had assumed a 0/8 control and cleared at $p = 0.026$. Separating 57% from 95% needs 16 rows a side ($p = 0.037$) and comfortably 20 ($p = 0.008$). This is a sharper instance of a lesson already recorded in §6.13's lemma-layer audit: a design must be sized against the measured between-draw spread of its own control, not against a control rate read off one draw. A stage-1 replication of the composition split is written out and frozen (`notes/2026-08-08-sarcasm-mood-floor-replication-preregistration.md`) — 40 rows, five targets, eight repeats split into two labelled blocks so the run measures its own noise — and is **recorded as not recommended for launch**: it would spend forty rows answering a question about the instrument rather than about tutoring. The line closes here on re-analysis rather than on a further run.

**What a stance comparison holds fixed, and what it does not.** A stance gate says whether a turn held its manner. It says nothing about whether two stances were asked for the same thing, and in this section they were not. Two columns above are unshared, and both are now printed beside every cross-stance count rather than left to the reader. First, the arms mandate different moves: the sarcastic contract asks for a concrete next move, the ironic one leaves the unmasking to the learner and adds an answerable test, the face-threat one asks for a minimal repair path, and the mild registers of the charisma cells mandate none. A difference between two fidelity counts is therefore a difference between two demands, not one demand met to different degrees — the same error as scoring the parent arm on cell 202's own contract, in a milder form that no gate check catches. Second, the instrument: the four sharp stances are scored by `config/rubrics/registers/irony-sarcasm.yaml`, while `charismatic` names no rubric of its own and falls back to the charisma rubric that scores this section's id-director cells. A charismatic score and a sarcastic score set in one column therefore come from two instruments on two scales, and cannot be differenced at all. Neither asymmetry is a defect in these runs, and neither is repairable by making the contracts alike — the moves *are* the stances. They are reasons to read the counts arm by arm. Both are now read from the register registry and rendered by every report that puts two stances side by side (`services/stancePayloadComparability.js`, ten tests), the scorer resolves a stance's instrument through the same function so no report can name a rubric the judge never opened, and an arm the registry cannot resolve fails the report closed rather than passing unchecked.

**What the fidelity counts count.** The stance gate's marker part looks for a fixed list of phrases plus the very cues the register registry hands the tutor, so it can see a manner only where the manner arrives as a listed phrase. A hand-marked set measures what that costs. Twenty tutor turns — five learner turns drawn from the grid run above, crossed with four writing conditions (ironic, sarcastic, face-threat, and a control naming no manner at all), one draw each on `codex.gpt-5.5`, with no registry cue phrase in any prompt (`services/registerEyeballSet.js`; a test parses the cue list out of the registry and fails if a prompt carries one). A reader marked all twenty blind to condition and before seeing any machine answer; a second reader on a different model family from the writer (`claude-sonnet-5`) then answered the same presence question, one manner at a time, verdict plus quoted evidence; the marker list answered it third. **Reading recovers the manner and the marker list does not.** The blind hand marks named the written manner on 19 of 20 turns and named none on all five controls. The blind model reader found the written manner on 15 of 15, declined on 14 of its 15 control judgments, and agreed with the hand marks on 50 of 60 judgments. The marker list found the written manner on 1 of 15, by way of "apparently" — one of the five cue phrases the registry hands the tutor. Its errors are mechanical and re-checkable rather than statistical: the ironic list fired on five turns and not one of them was written ironic, three of those by matching the string `so the`; two control turns carrying no manner scored 100 and passed, while every ironic turn scored 65 or 30 and failed; and the face-threat list matched none of the five face-threat turns, missing "protecting you from" on the pronoun and "That is an avoidance" on the word boundary.

Two consequences follow for the counts above and one for the designs. Every faithful count in this section — 5/15, 18/45, 7/15 against 8/15 — counts whether a tutor turn used a listed phrase, in these runs mostly a cue phrase it had been handed. Under that description the counts stand and the cue-repair result is exactly what it says: a compliance repair. What they do not license is the reading that the excluded rows dropped the manner, since on this set the marker fires on turns not written in the manner and misses the ones that were. The assigned-arm estimand is untouched, never having consulted the gate; the faithful-arm estimand was unlicensed rather than withdrawn, pending a re-reading of the stored rows — a re-reading that can cover the ironic and sarcastic arms only, for the reason given below. That re-reading has since run, and is reported in the next paragraph. The design consequence is that the two sharp registers are one category by reading. Both readers put exactly the same ten turns inside "ironic or sarcastic" — the five written ironic and the five written sarcastic — and neither placed a face-threat or a control turn there: 10 of 10 with no false alarm, twice, independently. Neither could split them. Every sarcastic turn was also read ironic, and the model reader read three of the five ironic turns as sarcastic besides. Sarcasm is here the sharper case of irony rather than its neighbour, so the ironic and sarcastic arms above overlap in extension and no result should turn on separating them; the arm-by-arm counts are reported with the overlap stated. Where a reading instrument is weakest is the finer category: the model reader called face threat on eleven turns where the hand marks called it on five, reading the naming of an avoidance into edged sarcasm. Scope is one writer model, one reader model, one draw per cell, twenty turns; the marker failures are deterministic and re-runnable by anyone, while the readers' agreement is a small check rather than a validated instrument. Artifacts: `tests/fixtures/register-eyeball-set/`.

**What a re-reading of the stored rows found.** Every stored tutor turn in an edged register was then read for manner: 62 turns across four runs (`eval-2026-08-05-87fe3664`, `eval-2026-08-06-4de45d05`, `eval-2026-08-07-45154bac`, `eval-2026-08-07-e3dffab2`), one merged edged question per turn, answered by a pinned Sonnet-class reader on a different model family from the writer (recorded as `codex.gpt-5.5`, actually `nvidia/nemotron-3-nano-30b-a3b`; a different family from the reader on either reading, so the independence the design wanted survives the correction), each answer cached under a hash of the prompt version, the reader and both turns (`services/registerMannerPresenceReader.js`). The reader found an edge in 24 of the 62 turns; six turns exceeded a three-minute call limit on the first pass and all six answered on a re-ask at ten minutes, so there are no unread rows. **The cue check and the reading name the same turn about two-thirds of the time either counts one.** The cue check passes 25 turns and the reading finds 24, and they agree on 16: it passes 9 turns carrying no edge and misses 8 that carry one. Per run and register — passed / of those edged / excluded yet edged — the grid run gives ironic 6/3/0 and sarcastic 8/6/2 of 15 rows each; cell 202 gives 4/3/5 of 15; the three-row ironic re-draw gives 0/0/0; the precondition follow-up gives 7/4/1 of 14. The two errors cost differently. Passing an unedged turn inflates a fidelity count, which is what this section reports; missing an edged turn only deflates one, because a reading cannot rescue a row the surface gate already excluded, so every faithful count here is a lower bound on manner presence. Two corrections to the grid run follow and must not be run together: deleting the pre-2.0 gate's hand-written phrase list takes 18/45 to 17/45, one face-threat row having passed on a word ("avoiding") that no registry cue handed it, and the reading then takes 17/45 to 12/45 (ironic 6→3, sarcastic 8→6, face-threat 3 unchanged). Face threat keeps all three rows because its question is never asked, so its count stays cue compliance and each arm now carries which of the two its faithful count is made of. Positive local outcomes on the 12 evidence rows: 9/12 overall, ironic 3/3, sarcastic 3/6, face-threat 3/3 — so the grid's earlier reading that ironic uniquely pairs faithful outcomes with faithful execution survives on a third of the rows it was drawn from, and no arm ordering should be read off $n \leq 6$. Cell 202's rows are reported beside the parent arm and not differenced against it: both counts now share a gate and a slice fold, but a contract only the treatment received is not a robustness check across arms. One limit on independence: where the reader says yes, the evidence it quotes is often the cue the tutor was handed ("wonderful", "Conveniently", "nice trick"), unlike on the hand-marked set, where no prompt carried a cue — what keeps the reading from restating the cue check is that they disagree on 9 rows in one direction and 8 in the other, which a reader tracking cue presence could not produce. Scope: one writer model, one reader model, one reading per turn, no second reader on this corpus; all simulated, non-human-facing. Artifacts: `exports/negative-register-manner-presence/`, `exports/negative-register-effect-grid/eval-2026-08-05-87fe3664.{md,json}`.

**Which models actually wrote these turns.** Every negative-register run in this section records `codex.gpt-5.5` in its `ego_model` column, and none of them called it. The id-director builds its ego and id requests from the cell block in `config/tutor-agents.yaml`; the run's command-line model overrides landed on the run configuration, which is what the column is written from, and never reached the engine. The dialogue logs settle it directly, since they record the provider and model of each call as it was made: per three-turn dialogue, three ego calls on `nvidia/nemotron-3-nano-30b-a3b`, three id calls and three agency-return-verifier calls on `moonshotai/kimi-k2.5`, with `codex.gpt-5.5` appearing only on the *learner* seat, which is configured elsewhere and ran as intended. This is exactly the pairing the project's standing rule forbids as a default, run for four days under the label of the strong stack. Four runs are affected and only these: `eval-2026-08-05-87fe3664` (cells 196–198, 45 rows), `eval-2026-08-06-4de45d05` (cell 202, 15), `eval-2026-08-07-e3dffab2` (cell 202, 14) and `eval-2026-08-07-45154bac` (cell 196, 3). The earlier July id-director runs match their labels; nothing in the code changed between them, so the July worktrees must have carried an edited cell block, and no `provider: codex` has ever been committed to that file. **What it costs is the stack, not the rows.** The turns were generated, stored, judged and read exactly as described — no count above changes, and the reading of the stored rows is untouched, its reader having been on a different model family from the writer under either name. What changes is what the counts are counts *of*: they are register fidelity on a weak open-weights pairing, and the standing rule holds such results stack-bounded until replicated. Read that way the section's headline is unaffected and one reading of it is closed off: canary-scale fidelity does not survive grid scale *on this stack*, and whether a stronger writer holds the manner more often is now an open question rather than a settled one. The hand-marked twenty-turn set and the twenty-row learner-turn probe are outside the correction — both call the CLI bridge directly, and both ran `codex.gpt-5.5` as recorded. Two defects were found and both are repaired with tests: the engine now applies a run's model overrides to the ego and id seats, and the weak-stack warning, which had skipped any run carrying a tutor-side override and so was silenced by the very flag that made these runs look strong, now reads the models a run will actually call.

**Does the sarcastic edge survive a strong writer?** One arm was bought back on the repaired engine to find out (`notes/2026-08-09-register-strong-stack-replication-preregistration.md`; plan SHA-256 `399d6188…955d`, printed by the dry run and operator-authorized on the commit that carried it): cell 197, the parent sarcastic arm, its cell block unchanged, against the same five controlled resistance targets × 3 repeats, `codex.gpt-5.5` on the ego and the id by an override that now reaches both seats, and everything else held — the same gate (`sarcastic` at `stance-gate/2.0`), the same adopting-turn fold, the same tutor and register judges, the same pinned reader on the same versioned question. Run `eval-2026-08-08-6021754f`, 15/15 rows, 15/15 tutor scores, 27/27 register slices, 15/15 read for manner with none unread; report COMPLETE. Which stack ran is itself a registered measure here, read off the dialogue traces rather than the model columns, because this run exists precisely because those two can disagree: 135 tutor-seat calls, all 135 on `codex/gpt-5.5`. **The strong writer**

**complies every time and holds the manner more often, but only the first of those separates.** Cue compliance is 15/15 against the parent arm's 8/15 (two-sided Fisher $p = 0.0063$); the reading calls 11/15 edged against the parent's 6/15 ($p = 0.1394$), which fifteen rows cannot tell from chance, so the pre-registered verdict — keyed to the reading, the registered primary — is no separation at this size, not agreement. Positive local outcomes on the faithful rows: 9/11. Two things follow. The weak pairing's shortfall was mostly a writer failing to use a cue phrase it had been handed, not evidence about the register: seven of its fifteen turns missed the cue and none of the strong writer's did. And the gap between saying the words and meaning them survives the stack change, since a writer that uses the cue in every turn still writes the manner into only eleven of them. *Post hoc, found by reading the four flat turns after the fact and not registered:* what separates them is a mock-compliment — praise in the words that the sentence then withdraws. Ten of the eleven edged turns carry one ("nice trick", "Conveniently", "Apparently", "Wonderful"); one of the four flat turns does ($p \approx 0.033$, post hoc). The flat turns state the cue and then teach straight: one opens by praising the learner's answer outright, one diagnoses without mocking, one gives the cue bare and proceeds to patient exposition. That the reader is not simply detecting the cue is visible in a pair of turns opening with the identical sentence, "Wonderful: the formula can sound like understanding while doing none of the work", of which it called one edged and one flat — it is judging the whole turn. Scope: one arm, one writer, one reader, one draw per cell, fifteen rows against fifteen; a screen for collapse rather than an estimate, cross-run and cross-stack by construction, and the stack change moves two tutor seats at once, so no part of the move can be pinned on the ego or the id alone. All simulated, non-human-facing. Artifacts: `exports/register-strong-stack-replication/eval-2026-08-08-6021754f.json`, `exports/negative-register-manner-presence/eval-2026-08-08-6021754f.json`.

**Does asking for the mock-compliment directly raise the reading?** The post-hoc observation above was made prospective under a frozen two-arm design (`notes/2026-08-09-register-mock-praise-preregistration.md`; plan SHA-256 `83b9ebe2…4a00`, printed by the dry run and operator-authorized on the commit that carried it). A new register, `sarcastic_mock_praise`, copies `sarcastic` and changes it as a package: the contract demands praise granted and withdrawn in the same breath, names diagnosis-without-merit as noncompliance, and drops the two non-praise cues from the cue family; cell 203 is cell 197's block with only the register changed, and the delivery path has no new code because the tutor's manner block composes from the registry. Two arms in one batch — control the plain `sarcastic` arm, treatment the asking one — 15 rows each on the same five targets × 3 repeats, `codex.gpt-5.5` on both tutor seats, the same judges, and the presence question deliberately unbumped so readings pool within the batch and against the stored run. Two limits were registered before the run: against a control fixed at 11/15 a 15-row arm cannot separate even at a perfect 15/15 (two-sided Fisher $p = 0.0996$), so the primary is a screen and the run's value rests on the manipulation check and the prospective device test; and the presence question's own gloss names the device ("praises what it is faulting"), so a treatment rise would show the reader can be satisfied on request, not that the turns teach better. Run `eval-2026-08-09-bb402d97`, 30/30 rows, 30/30 tutor scores, 50/50 register slices, 30/30 read with none unread; provenance 270/270 tutor-seat calls on `codex/gpt-5.5`; report COMPLETE. **The question dissolves, because the control**

**moved.** The plain arm, re-drawn one day after the run above, is read 14/15 edged and delivers a mock-compliment in 14 of 15 adopting turns unprompted — against its own 11/15 and 11/15 the day before — so the stored shortfall was between-draw noise rather than a floor for the ask to raise. The treatment is read 15/15 edged with praise in 15/15; the within-batch primary is 15/15 against 14/15 ($p = 1.0$), verdict no separation at this size, and the cross-run secondary lands at the pre-computed $p = 0.0996$. The registered device test is the one measure that moves: pooled over both arms, all 29 turns that grant praise in words are read edged and the single turn without praise is read flat ($p = 0.033$), reproducing the stored table (10/11 against 1/4) — on one bare turn, where the design had hoped the unasked control would supply several, since the plain register's cue family already delivers the device near ceiling on this writer. Asking costs nothing: tutor v2.2 means 66.3 against 66.7. Where this leaves the arc: across the 45 strong-stack turns of the two batches, praise-granted-and-withdrawn and the edge reading never come apart (39/40 with the device read edged, 1/5 without), so on this stack the withdrawn compliment is in practice how the writer realizes sarcasm and how this reader recognizes it; but no causal claim is licensed — one reader, whose gloss names the device — and naming the device in the contract adds nothing a strong writer was not already doing. Scope: two arms, one batch against one stored batch, one writer, one reader; simulated, non-human-facing. Artifacts: `exports/register-mock-praise-probe/eval-2026-08-09-bb402d97.json`, `exports/negative-register-manner-presence/eval-2026-08-09-bb402d97.json`.

**Cross-judge sanity check.** All cell-101-family rows were re-judged under `openrouter.gpt-5.2` to test judge-model robustness. On the v2.2 rubric ($n = 27$ paired rows per cell), GPT systematically scores 10–15 points lower than Sonnet on the high-scoring cells (c103, c104, c105) and slightly higher on the low-scoring cells. The rank-order is preserved under both judges (c104 > c103 > c105 > c106 > c102 > c101); the absolute magnitude of c104's lift over c101 baseline shrinks from +45 (Sonnet) to +27 (GPT). On the charisma rubric ($n = 56$ paired rows; remainder credit-blocked), all per-cell deltas fall within $±$3 points and rankings are identical under both judges. The architectural ordering survives the judge swap; magnitudes are judge-conditional and should be reported with the judge identifier when cited. Full breakdown in `docs/cell-100-cross-judge-sanity-check.md`.

**Rubric-curriculum alignment caveat.** A specific methodological concern arises because course 901 (AI literacy) is grounded in the same source paper that grounds the charisma rubric. A within-pilot ablation (`docs/cell-100-methods-note.md`) runs the c101 cell on the aura learner messages routed against the *non-aligned* 601 history-tech curriculum. Result: charisma 72.5 vs 75.8 on the aligned curriculum (3.3-point drop, well within sample variance). The rubric-curriculum alignment is not a confound that excludes aligned-cell results from primary architectural claims, but the disclosure is recorded for transparency.

**Replication, robustness, and orientation.** The id-director extension differs from the paper's mechanism-tracing core in scope: single judge primarily (Sonnet 4.6, with GPT-5.2 cross-judge sanity check on $n = 27$ paired rows per cell), single model family (Kimi K2.5 id $\times$ Nemotron-3-nano ego), per-cell $n$ ranging from 27 (cell 107 closing pilot) through 54 (cells 102/106) to 79–81 (cells 101/103/104/105), and a different theoretical orientation (Weber rather than Hegel). It is presented here as architectural exploration rather than as part of the three-mechanism model evaluated in §6.1–6.6. The relevant claim is *robustness of architectural ranking* across scenario sets, judges, and curricula, not magnitude estimates. Computational and credit limits (documented inline) prevented the full multi-judge × multi-model replication that the paper's main study uses. See §8.9 for explicit scoping.

**Reproducibility.** Pilot run IDs (all on branch `experiment/cell-100-id-charisma`):

- `eval-2026-04-26-220628d4` (601 history-tech original); `eval-2026-04-26-4df177e6` (701); `eval-2026-04-26-0d15c1b5` (901)
- `eval-2026-04-27-23beaa63` (601 cells 103/104); `eval-2026-04-27-3115ab29` (701); `eval-2026-04-27-2d5c20ca` (901)
- `eval-2026-04-27-69167b53` (601 cells 105/106); `eval-2026-04-27-2efec307` (701); `eval-2026-04-27-ce22e888` (901)
- `eval-2026-04-27-dce2a0f1` / `1f192b31` / `18dbb95d` (replication scenarios across 3 curricula)
- `eval-2026-04-29-499ab3d2` / `e35c66ae` / `be822796` (c108 full-N follow-up across 3 curricula, $n = 27$)
- `eval-2026-04-26-759e1a02` (Item 1 curriculum-swap ablation)
- `eval-2026-04-28-3620e0f8` / `a638111d` / `d6f5fee2` (cell 107 witness-exemplar smoke, codex/design/tools)
- `eval-2026-04-28-ee339a66` / `84a42a60` / `5606193c` (cell 107 closing pilot, lecture+phaedrus / fairness+team / authentic+consumption)
- `eval-2026-04-28-10216f9f` (cells 108/109 lever-interaction pilot, n=11 across 2 scenarios)

Five internal documents (`docs/cell-100-pilot-findings.md`, `docs/cell-100-methods-note.md`, `docs/cell-100-pilot-findings-addendum.md`, `docs/cell-100-cross-judge-sanity-check.md`, `docs/cell-100-replication-findings.md`) carry the full audit trail; this section summarises only headline findings. Per-row JSON in `evaluation_results.tutor_charisma_scores`, `evaluation_results.scores_with_reasoning`, and `evaluation_results.id_construction_trace`. Note: the cell IDs in this section are the post-rebase identifiers (101–107); pre-rebase historical evaluation_results rows used the original `cell_100_*` profile names (the rebase commit `02fe11e` shifted them to avoid a collision with main's `cell_100`).

---

### 8.9 Scope of the Id-Director Extension

The id-director family findings (§6.7, cells 101–107) differ from the paper's mechanism-tracing core in scope along four axes. *Single-judge primary scoring*: v2.2 scoring used `claude-code/sonnet` exclusively, with `openrouter.gpt-5.2` as a cross-judge sanity check on $n = 27$ paired rows per cell — far short of the three-judge $\times \geq 144$-row matrix supporting the paper's core mechanism findings (§5.8). The cross-judge check confirms architectural rank-order but compresses absolute magnitudes by 10–15 v2.2 points; published numbers should specify the judge. *Single model family*: id is Kimi K2.5 across all id-director cells; ego is Nemotron-3-nano. The paper's core findings are validated across DeepSeek V3.2, Haiku 4.5, and Gemini Flash 3.0 (§5.8); the id-director extension is not. Whether the architectural ranking generalises to other id $\times$ ego model pairings is open. *Per-cell N is heterogeneous and partly below the paper's core floor*: closing N is $n = 79$–81 for the four central cells (101, 103, 104, 105), $n = 54$ for cells 102 and 106 (two of three scenario types covered), and $n = 27$ for cell 107 (the most exploratory closing pilot). Three of seven cells therefore meet or exceed the paper's core $n \geq 63$ per dialogue-condition floor; one cell (c107) sits substantially below it and is reported as a pilot-scale rather than confirmatory result. Variance estimates on the lower-N cells should be read accordingly. *Different theoretical orientation*: the charisma rubric grounds in Weberian charismatic authority (Weber, 1978) rather than Hegelian recognition. The two are correlated but not redundant; the §6.7 results do not displace recognition theory but document an architectural variant in which a complementary frame is also load-bearing. The §6.7 claims are framed as architectural exploration rather than as additional mechanisms in the three-mechanism model. Where the id-director extension introduces methodological novelty (the prompt-budget non-monotonicity, the rubric-curriculum alignment ablation, the cross-judge magnitude-compression observation), the contribution is descriptive and supplementary to the paper's main argument; full audit trail is in the five `docs/cell-100-*.md` documents and the dialogue-log JSON for each pilot run.

The desire-for-recognition follow-up in §6.7 is narrower still. It is an iterative design slice, not a confirmatory matrix: one run per listed profile-scenario row in the cells 159–169 authority-refusal search; selected static-profile diagnostics in cells 170–179; then a local resistance-breakthrough router line in cells 180–195. The first result is sufficient for a paper-side design finding ("accountable-bid charisma is the first clean local pass under authority refusal") but not sufficient for a general claim about charismatic tutoring. The second result shows why a static-cell ladder is the wrong target: transfer demand, authority refusal, plain-language request, simplification follow-up, and resistance after scaffolding are different engagement states. The third result shows that an engagement-register router can work locally under the Codex/Claude stack (cell 193: 25/26 positive local mechanism outcomes across the combined eligible evidence) but also supplies the boundary: GLM 5.2 generation does not reproduce the mechanism, and even provider/runtime reasoning controls that produce clean DB validation leave the local breakthrough outcome at 0/5 on the five-signal check. A completed six-arm role-isolation grid now identifies the most plausible boundary as dynamic-learner completion and target drift rather than tutor/id public-register production: the GLM tutor/id with Codex learner arm remains locally usable (8/10 positive outcomes), the GLM dynamic-learner arms lose post-turn completion or target stability, and both scripted tutor controls route the public register with fixed resistant turns. The separate question-flood gate promotes cell 192 only for that subtype (2/2 answer-first usable commitments, 0 reopened or residual floods), not for the broader five-signal mechanism or GLM stack. To upgrade the finding after that pivot, the router would need a frozen grid spanning authority-withheld, status-challenge, partial-uptake, ordinary conceptual, vulnerability/persona-shift, transfer, plain-language, and instruction-to-engagement-switch scenarios; $\geq 10$ repeats per profile-scenario as a pilot floor and preferably $\geq 24$ for stable variance estimates; frozen comparators including cell 163, cell 104/107-style id-director baselines, the default tutor, and a no-router id-director baseline; at least two judges for both v2.2 and charisma scoring; model-pair variation across Codex/Claude, non-CLI hosted models, and smaller ego models; and a human-coded or human-learner subset focused on whether learners experience the tutor's authority as defeasible rather than merely polished. Without those extensions, cells 169, 193, and the cell-192 question-flood subtype should be cited as exploratory design evidence motivating engagement-register routing, not as a generalizable effect.

The negative-register extension adds a further scope condition: arm assignment is not treatment fidelity. A cell can be configured as `sarcastic_challenge` while the generated tutor turn remains warm, ironic, or only superficially edgy. The hand-authored corrosive-exemplar check in §6.7 shows that the scoring instruments can detect corrosive sarcasm when fixed examples actually contain it, so the immediate limitation is not global judge blindness but generated-register fidelity. A fresh stance-gated smoke confirmed the practical scale of that limitation before repair: only 5/15 assigned negative-register rows counted as faithful arm evidence, while 10/15 were excluded as weak or warm-in-costume noncompliance and none were invalid person-attack violations. Later cue-repair canaries then showed that explicit stance-fidelity cues can restore treatment fidelity on sampled targets: 9/9 faithful, 0 invalid across boredom/irrelevance/rote-parroting, followed by 6/6 faithful, 0 invalid on held-out frustration/question-flood. Across the two post-repair canaries, all five resistance targets have now been sampled without gate exclusions or invalid person-attack violations; this is still an implementation check rather than an effect estimate. Negative-register claims must therefore report both assigned-arm and faithful-arm estimands, and rows should contribute to a negative-register effect estimate only after the adopting tutor turn is classified as faithfully instantiating the assigned register contract. A related scope condition concerns the judge rather than the tutor: the register rubric's own scores were originally produced by `openrouter.gpt-mini`, later shown to be numerically non-discriminating on this rubric (§6.7); every register-rubric conclusion for cells 196–198 now rests on a re-score under a sonnet-class judge (`claude-code.sonnet-5`), and the archived gpt-mini register-rubric figures are void and must not be cited. Sonnet-class judging is the standing requirement for the register rubric from the first row of any future negative-register run. The pre-registered 45-row effect grid (§6.7, run `eval-2026-08-05-87fe3664`) sharpens the fidelity scope condition with scale evidence: the cue-repair fidelity that held 15/15 across the two canaries fell to 18/45 under the same gate at grid scale — 17/45 once that gate's phrase list was deleted, and 12/45 once the rows were read for manner (§6.7) — so canary-scale fidelity licenses no scaled-fidelity assumption on this stack, and every effect reading exists only as the assigned/faithful pair — exploratory at three repeats per cell, simulated-only. The determinate-negation follow-up (§6.7, cell 202, run `eval-2026-08-06-4de45d05`) adds two scope conditions, one about measurement and one about the contract. The measurement one is that **a stance count is not a measurement until it names both its gate and its slice fold**. This paragraph first read the follow-up as lowering fidelity to 6/15 from the parent arm's 8/15; those counts differed in the gate that produced them *and* in the unit of text the gate was applied to, and holding both fixed leaves fidelity flat at 7/15 against 8/15 (corrected at v3.0.269). Register verdicts now carry the identity and version of the gate that produced them, and the differencing code refuses to subtract verdicts that disagree on either, because a count read off a mismatched pair is not weak evidence but no evidence. The substantive one is that **a contract binds only where its precondition holds**: requiring each sarcastic turn to name the learner claim it negates presupposes that the learner has advanced a claim, and against boredom and frustration — where the resistance is a stance rather than an assertion

— there is often nothing determinate to name, which is where the manner went missing (0/6 faithful, against 5/6 in the parent arm). A register contract should therefore state the conditions under which it is satisfiable. *[**This substantive condition is withdrawn as an empirical claim and retained only as a design maxim.** The 0/6 it rests on is a single draw; the same cell on the same two mood scenarios, byte-identical in configuration, gave 4/7 one day later (§6.7, $p = 0.070$). Stating a contract's satisfiability conditions remains good practice, but this section reports no measured case of a contract failing for want of its precondition. The measurement condition above — name your gate and your fold — is untouched, resting on a comparison of two counts rather than on the level of either.]* A third condition follows from the same re-analysis and is about designs rather than instruments: **a control rate read off one draw is not a base rate.** The precondition follow-up was sized against an assumed 0/8 control, at which it cleared $p = 0.026$; at the 57% control the replication actually measures, its best possible result reaches only $p = 0.077$, so no outcome it could have produced was readable. The lemma-layer audit records the same hazard from the other side — identical cells at $n = 6$ swinging by two outcomes across seed draws. A design must be sized against the measured between-draw spread of its own control, which means measuring that spread before, or inside, the design that depends on it. And a checkable content requirement is not a lever on manner: the added conjunct turned out to be nearly independent of whether a turn held its register at all (§6.7), because it can be met in an entirely earnest voice. That follow-up's negation-recovery measure is reported as underpowered rather than null: one recovery event across six faithful rows can neither support nor refute co-movement with local outcomes, so no manner-as-content mechanism claim is licensed in either direction, and the parent grid's sarcasm result stands as a manner-only effect. The precondition follow-up (§6.7, run `eval-2026-08-07-e3dffab2`) adds a third condition, again about the instrument: **a gate must state which of its components are necessary, not leave necessity to its weights.** The register marker was necessary on the plain gate only because $100 - 35$ fell below the faithful band; re-weighting it to 25 for the determinate variant repealed that without anyone writing a line of it down, and the variant then labelled marker-less earnest turns faithful. A measure named "held the manner" spent two runs not looking at the manner. Necessity is now written as a rule on both gates rather than inferred from arithmetic, and the counts that moved are corrected in §6.7. The same follow-up retires a second habit: **a treatment-specific gate is not a robustness check across arms.** Scoring the parent sarcastic arm under `sarcastic_determinate` grades a control on the contract only the treatment received, so its agreement or disagreement with the plain gate says nothing about whether the plain-gate reading is robust. A fourth condition, from the hand-marked set of §6.7, is about what a gate of this kind can see at all: **a phrase list cannot measure a manner defined by the gap between what a sentence says and what it means.** The marker part matches a fixed list plus the cues the registry hands the tutor, and on twenty turns written to order it found the assigned manner once in fifteen while passing two control turns that named no manner. Every faithful count in this arc therefore reports phrase compliance rather than manner presence, and the exclusions cannot be read as the manner failing to arrive; the assigned-arm estimands are unaffected, and the faithful-arm ones awaited a re-reading of the stored rows. That re-reading reaches two of the three arms. It has since run over all 62 stored rows in edged registers (§6.7), and its result is a sixth condition of the same kind: **a surface check and a reading of the same**

**turns are different measures, and how far they overlap is a number to report rather than assume.** The two name the same turn 16 times out of the 25 the check passes and the 24 the reading finds; the check passes 9 turns carrying no edge and misses 8 that carry one, which takes the grid run's faithful count from 17/45 to 12/45 and leaves every such count a lower bound, since a reading cannot rescue a row the surface gate already excluded. The same set that shows reading recovering what matching cannot also shows the model reader over-attributing the finer category — face threat on eleven turns against five written, and once on a control turn carrying no manner — so there is no validated presence question for that arm, and its faithful count stays a count of cue compliance whatever a re-reading of the other two returns. The same set licenses the positive half of the condition: two blind readers, one of them a model on a different family from the writer, separated the ten edged turns from the ten unedged ones with no false alarm on either side, so presence is measurable by reading where it is not by matching — and the gate's other parts, which check speech acts that do live in the surface (a named target, a next move, a repair path), are untouched by any of this. A fifth follows from the same set and is about the designs: **the ironic and sarcastic arms are overlapping extensions, not alternatives.** Neither reader could split them, every sarcastic turn being read ironic as well, so no claim in this arc rests on telling the two apart. A seventh condition is about the stack rather than the instrument, and it bounds every count in this arc: **the four August negative-register runs ran on a weak open-weights pairing, whatever their model column says.** The id-director assembles its ego and id requests from the cell YAML, so the runs' command-line model overrides never reached it; the rows record `codex.gpt-5.5` while the dialogue logs show `nvidia/nemotron-3-nano-30b-a3b` on the ego and `moonshotai/kimi-k2.5` on the id and the reviewer, with `codex.gpt-5.5` reaching only the learner (§6.7). No count moves — the turns were generated, judged and read as described — but the project's standing rule holds results on that pairing stack-bounded until replicated, so every fidelity figure for cells 196–198 and 202 is a figure about that stack. The engine and the weak-stack warning are both repaired, and one arm has since been re-run on the repaired engine to answer the substantive question (§6.7, cell 197, run `eval-2026-08-08-6021754f`, 15 rows, provenance checked off the dialogue traces at 135/135 tutor calls on `codex.gpt-5.5`). It answers half of it. The strong writer used the handed cue in every turn against the weak pairing's 8 of 15 ($p = 0.0063$), so **most of the fall from canary fidelity to grid fidelity was a weak writer failing to follow an instruction, not scale.** The manner itself is a different matter: the reading calls 11 of 15 edged against 6 of 15, which fifteen rows cannot separate from chance ($p = 0.14$), and that is reported as no separation at this size rather than as agreement. So the edge does not collapse on a strong writer, the gap between using the cue and writing the manner survives the stack change, and the arm-level fidelity counts for cells 196–198 and 202 remain bounded to the pairing that produced them. A pre-registered two-arm follow-up on the repaired engine (§6.7, cells 197/203, run `eval-2026-08-09-bb402d97`) adds a base-rate caveat to that half-answer: the same plain arm re-drawn one day later is read 14/15 edged, so the 11/15 was a draw rather than a rate, and the residual cue-versus-manner gap on this writer sits inside between-draw noise; what is stable across the two batches is the device-level regularity — 39 of the 40 strong-stack turns that grant praise in words and withdraw it are read edged, against 1 of the 5 that do not — registered prospectively in the follow-up ($p = 0.033$ in that batch alone), with no causal reading licensed because the presence question's own gloss names the device. Cells 196–198 and 202 remain a measurement-development addendum plus two exploratory simulated grids, not evidence about the pedagogical value or safety of sarcasm.

A final scope condition, terminal for the negative-register arc as a whole, is about where a register can act at all. Five designs in this programme have varied the manner of the tutor's speech while its moves stayed fixed or effectively fixed, and none produced a resolved positive outcome: the §6.13.19 register router (conduct measurably changed, outcomes identical at 16/20 grounded per arm); the adaptive register-switching arm (no downstream learner improvement; workplan card `adaptive-register-switching`); the 312-row edged-register outcome block (unresolved at 48% power behind a reader-side confound; card `edged-register-outcome-study`); the §6.27 v5 warm-manner contrast on a bored learner (1/18 against 4/17, $p = 0.18$); and §6.29, where a scripted core froze every teaching move and warm against sarcastic produced an operator-accepted descriptive null on the proof-DAG in three learner characters. Against these stand the arcs in which the negative or challenging element was a *move* — a timed challenge (§6.25), a warrant demand on an overconfident learner (§6.26), a delivered re-engagement move on a resistant one (§6.28) — which produced this programme's registered positives. The boundary this draws binds every register claim in the paper: on these stacks, with simulated learners, manner divorced from moves has not produced a resolved positive outcome, so register effects should be sought, and will be claimed, only where the register is permitted to change what the tutor does rather than only how it sounds. This is design guidance, not a theorem: the five non-positive or inconclusive results are stack- and harness-bounded, and the manner-through-moves channel is untested by construction, since every one of these designs deliberately froze it.

---

**Neighbouring modules:** Module 5 — Can Productive Tension Improve Teaching?; Module 1 — Meeting the Learner Raises the Floor.

Fazeli, S. et al. (2023). *Visual-verbal video analysis (VVVA): A multilayered qualitative framework for short-form video.* Method note.

Weber, M. (1978). *Economy and society: An outline of interpretive sociology* (G. Roth & C. Wittich, Eds.). University of California Press.